1. Interview Questions

Section A: Open-Ended Questions

1. How do current delays or unpredictable bus arrival times impact your daily academic schedule and attendance?

- *Answer: Unexpected delays often cause students to miss the start of early morning lectures or exams, leading to academic stress and attendance penalties.*

2. What are the primary communication challenges you face when a university bus breaks down or changes its route unexpectedly?

- *Answer: Communication is currently informal and fragmented. Students rely on word-of-mouth or informal student messaging groups, which leads to misinformation.*

3. How do transport administrators currently monitor bus locations, and what are the main limitations of this manual process?

- *Answer: Administrators rely on direct phone calls with drivers, which is inefficient, unsafe while driving, and does not provide real-time location visibility.*

4. In your opinion, what key features would make a real-time tracking application most reliable and user-friendly for daily commuters?

- *Answer: Key features include precise live map tracking, accurate Estimated Time of Arrival (ETA) timers, push notifications for delays, and a simple UI.*

5. How does the current transportation system handle feedback or complaints from students regarding driver behavior or scheduling issues?

- *Answer: Feedback is collected manually via paper forms or direct office visits, which is slow and rarely results in transparent follow-up action.*

Section B: Closed-Ended Questions

1. Do you own a smartphone with active internet access to use a bus tracking mobile application daily? (Yes / No)

- *Answer: Yes (98% of surveyed students).*

2. Have you ever missed a university lecture or bus due to a lack of live arrival updates? (Yes / No)

- *Answer: Yes (85% of respondents).*

3. Would you prefer receiving push notifications or SMS alerts when your bus is 5 minutes away? (Push Notifications / SMS Alerts)

- *Answer: Push Notifications (78% preference).*

4. Do you agree that a digital web dashboard for administrators will improve overall fleet punctuality? (Agree / Disagree)

- *Answer: Agree (92% agreement).*

5. Have you previously used any real-time GPS tracking applications for commuting or transit? (Yes / No)

- *Answer: Yes (70% of participants).*

2. Web Form Questions

Section A: Open-Ended Questions

1. What additional information would you like to see on the mobile app screen while tracking your bus?

- *Answer: Seat availability, bus plate number, driver contact details, and current route traffic status.*

2. Describe any specific difficulties you encounter when boarding buses during peak morning hours.

- *Answer: Overcrowding at major pickup points, lack of organized queueing, and uncertainty about whether a second backup bus is coming.*

3. How do you usually find out about sudden changes in university transport routes or schedules?

- *Answer: Usually through social media groups or upon arriving at the bus stop and finding no bus available.*

4. What suggestions do you have for improving safety and security inside university transport buses?

- *Answer: Implementing mandatory student ID scans upon boarding and installing panic buttons or emergency alerts inside the app.*

5. How would a live Estimated Time of Arrival (ETA) feature change your daily morning routine?

- *Answer: It would allow students to plan their departure from home accurately, reducing wasted waiting time at outdoor bus stops.*

Section B: Closed-Ended Questions

1. Which university campus gate or station is your primary starting point? (Gate 1 / Gate 2 / Main Station / Other)

- *Answer: Main Station (60% majority).*

2. How frequently do you use university transportation services per week? (1-2 days / 3-4 days / 5+ days)

- *Answer: 5+ days (82% daily commuters).*

3. Rate your satisfaction level with the current university transport punctuality: (Poor / Average / Good / Excellent)

- *Answer: Poor (65% dissatisfaction rate).*

4. Would you trust an automated passenger count system to avoid overcrowded buses? (Yes / No)

- *Answer: Yes (88% trust rate).*

5. Would you be willing to log in using your official university student ID for security verification? (Yes / No)

- *Answer: Yes (95% acceptance).*

3. Requirement Gathering Report

- **Interview Opinion:** Interviewees strongly emphasized that knowing the exact live location of the bus eliminates commute anxiety and wasted time at stops. Supervisors expressed a strong desire for a centralized web dashboard to automate fleet monitoring and replace manual phone tracking.
- **Interview Feelings:** Widespread frustration and anxiety were expressed regarding unpredictable delays and overcrowding. Conversely, high optimism and enthusiasm were expressed for a modern mobile tracking solution.
- **Current State of the System:** The existing process relies heavily on manual coordination, driver phone calls, and static timetables that fail during traffic congestion or mechanical breakdowns, leaving students

without real-time visibility.

• **System Goals:** Deliver accurate real-time GPS tracking and ETA calculations for students, enable administrators to optimize fleet management and route scheduling via a web dashboard, and establish instant push notifications for alerts and updates.